PF233-HB2: HB2 grip board: Hall trigger, 4-position dial divider, signal cord

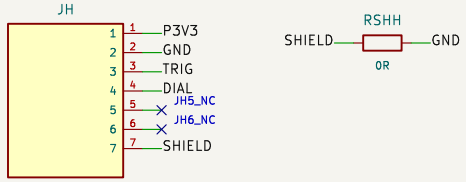
How to read: parts sit in functional blocks (dashed frames). Every pin ends in a short wire with its NET NAME; equal names are connected (also between blocks). A cross = intentionally unconnected pin.

Diode symbols point anode to cathode. Pin names are those of the library; items marked VERIFY in docs/ASSUMPTIONS.md are unverified. MODELED design: not built, not bench-tested.

1 Signal cord to the head

JH: +3V3 (P3V3), GND, TRIG, DIAL, two unused, SHLD.

RSHH (0 R) bonds the signal-cord shield to GND at the handle.

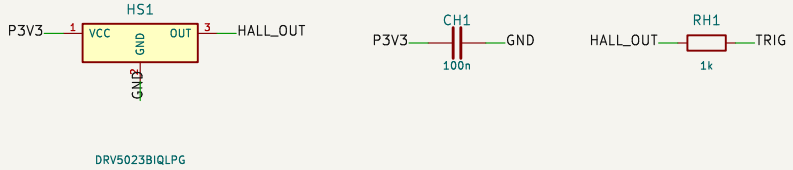


Signal cord to the head: +3V3 GND TRIG DIAL

2 Hall trigger

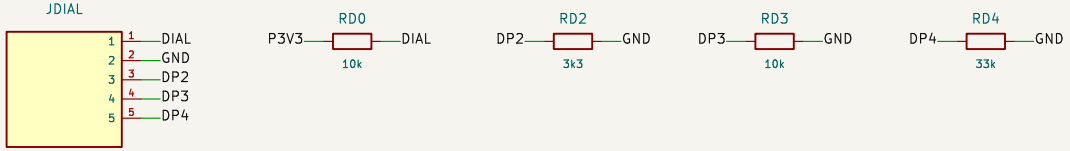
HS1 (digital Hall switch, open drain): magnet near at rest -> TRIG low.

RH1 1 k in series to the cord; CH1 100 nF decoupling.



3 Dial (4-position rotary)

RD0 10 k pull-up from P3V3 to DIAL (= JDIAL COM). Positions: P1 straight to GND; P2 / P3 / P4 through RD2 3k3 / RD3 10 k / RD4 33 k to GND, so each position gives a different divider level.



Dial: COM P1 P2 P3 P4 (4-pos rotary, top hou

Generated: src/pcb/sch_func.py; checked against the netlist.
MODELED design: not built, not bench-tested.
HB2 grip board: Hall trigger, 4-position dial divider, signal cord
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Title: PF233-HB2 functional schematic

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